EEEN 133 HW 2

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1. $y(n)=0$ for $n<0$

impulse response

a. $y(n) = x(n) - 2x(n-2) + x(n-4)$

$$\boxed{h(n) = \delta(n) - 2\delta(n-2) + \delta(n-4)}$$

n	$x(n)$	$-2x(n-2)$	$x(n-4)$	$y(n)$
-1	0	0	0	0
0	1	0	0	1
1	0	0	0	0
2	0	-2	0	-2
3	0	0	0	0
4	0	0	1	1

b.

$$\boxed{h(n) = \frac{1}{5}\sum_{k=0}^{4} \delta(n-k)}$$

c. $y(n) = x(n) + .6y(n-1)$

$$\boxed{h(n) = .6^n u(n)}$$

	$x(n)$	$.6y(n-1)$	$y(n)$
-1	0	0	0
0	1	0	1
1	0	.6	.6
2	0	.36	.36
3	0	.216	.216

d. $y(n) = x(n) - .4y(n-1)$

$$\boxed{h(n) = (-.4)^n u(n)}$$

	$x(n)$	$-.4y(n-1)$	$y(n)$
-1	0	0	0
0	1	0	1
1	0	-.4	-.4
2	0	.16	.16
3	0	-.064	-.064

2.

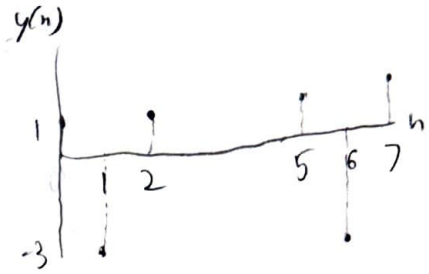
	linear	Causal	Stable	Time-inv	why?
a. $y(n) = c_1 x(n) + c_2$	X	✓	✓	✓	The const (c_2) breaks linearity
b. $y(n) = x(n^3)$	✓	X	✓	X	future input and index is not linear
c. $y(n) = n^3 x(n)$	✓	✓	X	X	n^3 is unbounded must not $(n-k)^3 \neq n^3$
d. $y(n) = \sum_{k=2}^{4} x(n-k)$	✓	✓	✓	✓	LTI
e. $y(n) = \sum_{k=-1}^{1} x(n-k)$	✓	X	✓	✓	future ($x(n+1)$) input

3. $h(n) = \delta(n) - 3\delta(n-1) + \delta(n-2)$

a. $x(n) = \delta(n) + \delta(n-5)$

$y(n) = h(n) + h(n-5)$

$$y(n) = \delta(n) - 3\delta(n-1) + \delta(n-2) + \delta(n-5) - 3\delta(n-6) + \delta(n-7)$$

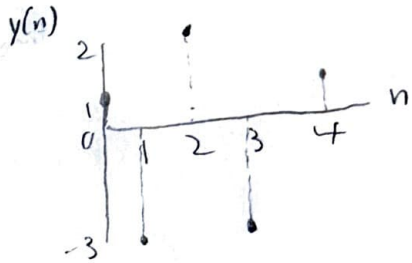


b. $x(n) = \delta(n) + \delta(n-2)$

$y(n) = \delta(n) - 3\delta(n-1) + \delta(n-2)$

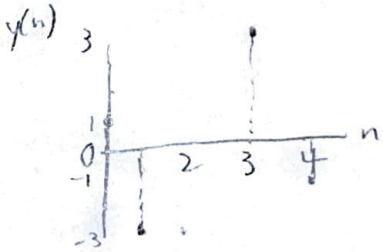
$+ \delta(n-2) - 3\delta(n-3) + \delta(n-4)$

$$y(n) = \delta(n) - 3\delta(n-1) + 2\delta(n-2) - 3\delta(n-3) + \delta(n-4)$$

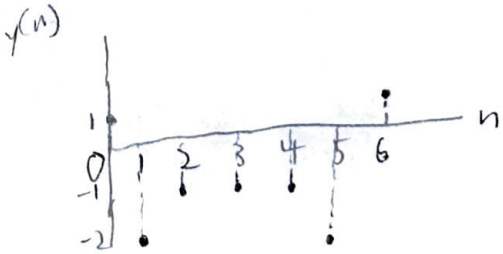


c. $x(n) = \delta(n) - \delta(n-2)$

$$y(n) = \delta(n) - 3\delta(n-1) + 3\delta(n-3) - \delta(n-4)$$



d. $x(n) = \delta(n) + \delta(n-1) + \dots \delta(n-4)$



n	$-3(n-1)$	$(n-2)$			
$\downarrow$	$(n-1)$	$-3(n-2)$	$(n-3)$		
1	$\downarrow$	$(n-2)$	$-3(n-3)$	$(n-4)$	
	-2	$\downarrow$	$(n-3)$	$-3(n-4)$	$(n-5)$
		-1	$\downarrow$	$(n-4)$	$-3(n-5)$ $(n-6)$
			-1	$\downarrow$	$\downarrow$
				-1	-2 1

$$y(n) = \delta(n) - 2\delta(n-1) - \delta(n-2) - \delta(n-3) - \delta(n-4) - 2\delta(n-5) + \delta(n-6)$$

4. series: $h_1 * h_2$, parallel: $h_1 + h_2$; $h_1(n) = \delta(n) - a\delta(n-1)$, $h_2(n) = \delta(n) + a\delta(n-1) + a^2\delta(n-2)$

a. $(\delta(n) - a\delta(n-1))(\delta(n) + a\delta(n-1) + a^2\delta(n-2))$

$$\delta(n) + a\delta(n-1) + a^2\delta(n-2)$$
$$-a\delta(n-1) - a^2\delta(n-2) - a^3(n-3)$$

$$\boxed{h_1(n) * h_2(n) = \delta(n) - a^3(n-3)}$$

b.

$$\boxed{h_1(n) + h_2(n) = 2\delta(n) + a^2\delta(n-2)}$$

5. $h(n) = \delta(n) + 4\delta(n-1) + \delta(n-2)$ a.

$$H(e^{jw}) = \sum_{k=-\infty}^{\infty} h(k) e^{-jwk} = \sum_{-\infty}^{\infty} (\delta(k) + 4\delta(k-1) + \delta(k-2)) e^{-jwk} = 1 + 4e^{-jw} + e^{-2jw}$$

$$e^{-jw}(e^{jw} + 4 + e^{-jw}) = \boxed{e^{-jw}(4 + 2\cos(w))}$$

b.

$$H(e^{jw}) = e^{-jw}(4 + 2\cos(w) = 1 + 4e^{-jw} + e^{-2jw} = 1 + 4(\cos(w) - j\sin(w)) + \cos(2w) - j\sin(2w)$$

real: $1 + 4\cos(w) + \cos(2w)$

imaginary: $-4j\sin(w) - j\sin(2w)$

c.

	real	imag.	mag	phase
$w=0$	6	0	6	0
$w=\frac{\pi}{4}$	$1+2\sqrt{2}$	$-1-2\sqrt{2}$	$4+\sqrt{2}$	$-\pi/4$
$w=\frac{\pi}{2}$	0	-4	4	$-\frac{\pi}{2}$
$w=\frac{3\pi}{4}$	$1-2\sqrt{2}$	$1-2\sqrt{2}$	$4-\sqrt{2}$	$-3\pi/4$
$w=\pi$	-2	0	2	$-\pi$

d. 10kHz

$f = \frac{w f_0}{2\pi}$ analog freq

$w=0$: 0 Hz
$w=\pi/4$: 1250 Hz
$w=\pi/2$: 2500 Hz
$w=3\pi/4$: 3750 Hz
$w=\pi$: 5000 Hz

frequencies' output amplitudes highest to lowest

0 Hz, 1250 Hz, 2500 Hz, 3750 Hz, 5000 Hz
(6) $(4+\sqrt{2})$ (4) $(4-\sqrt{2})$ (2)

e. 50 Hz

$w=0$; 0 Hz
$w=\pi/4$; 6.25 Hz
$w=\pi/2$; 12.5 Hz
$w=3\pi/4$; 18.75 Hz
$w=\pi$; 25 Hz

Same thing. The lower the freq (up to 0), the higher the amplitude.

6. $h(n) = \delta(n) - 1.5\delta(n-1) + \delta(n-2)$

a. $H(e^{jw}) = 1 - 1.5e^{-jw} + e^{-2jw} = e^{-jw}(2\cos(w) - 1.5)$

b. imaginary: $-1.5j\sin(w) - j\sin(2w)$ real: $1 - 1.5\cos(w) + \cos(2w)$

c.

	real	imag	mag	phase
$w=0$	$\frac{1}{2}$	0	$\frac{1}{2}$	0
$w=\pi/4$	$1-\frac{3\sqrt{2}}{4}$	$\frac{3\sqrt{2}}{4}-1$	$\approx .086$	$3\pi/4$
$w=\pi/2$	0	$\frac{3}{2}$	$\frac{3}{2}$	$\pi/2$
$w=3\pi/4$	$1+\frac{3\sqrt{2}}{4}$	$\frac{3\sqrt{2}}{4}+1$	2.914	$\pi/4$
$w=\pi$	$\frac{7}{2}$	0	$\frac{7}{2}$	0

d.

	magnitude response
$w=0$: 0 Hz	$\frac{1}{2}$
$w=\pi/4$: 1250 Hz	.086
$w=\pi/2$: 2500 Hz	$\frac{3}{2}$
$w=3\pi/4$: 3750 Hz	2.914
$w=\pi$: 5000 Hz	$\frac{7}{2}$

e.

$w=0$; 0 Hz	$\frac{1}{2}$
$w=\pi/4$; 6.25 Hz	.086
$w=\pi/2$; 12.5 Hz	3/2
$w=3\pi/4$; 18.75 Hz	2.914
$w=\pi$; 25 Hz	7/2

Unlike last problem, as the freqs get larger, the amplitude gets bigger EXCEPT for 6.25 Hz and 1250 Hz. These freqs are dampened and "filtered" out essentially.